

SOLUTIONS TO SOME OF THE QUESTIONS ON THE STATISTICS APPLIED MASTERS COMPREHENSIVE EXAM

These are Solutions that have been found to some of the questions that were asked since 2008.

Please Note that in some cases what is presented here may be the initial questions and solutions that were submitted when the exam was prepared. It is possible that the questions (as well as the solutions that were used for grading) for the exams which were given may have changed slightly to end up being different.

Also Note that these answers may not be the only way to address the question, in some cases there may be other solutions that would get partial or even full credit.

Also in Blue the **COMPETENCIES** reflect the areas of knowledge that the questions are covering.

FALL 2018

1. A random sample of 20 adult Busch Bears is obtained from a forest in New Jersey. The following stem & leaf gives the weights in pounds for these animals

1 | 2789
2 | 12346679
3 | 1358
4 | 0112

COMPETENCIES = Descriptive Statistics, Binomial Distribution Properties

a) Compute the median weight in the sample_____

$$(26 + 27) / 2 = 26.5$$

b) Compute the interquartile range of weight in the sample_____

$$(21 + 22) / 2 = 20.5 \text{ to } (35 + 38) / 2 = 36.5$$

c) The median weight of an adult Rutgers-Wombat is known to be 39 pounds. Test the null hypothesis that the median weight of an adult Busch bear is heavier than this against the alternative that it is not at $P = 0.001$.

Best answer

$$\left[\binom{20}{0} + \binom{20}{1} + \binom{20}{2} + \binom{20}{3} + \binom{20}{4} \right] * 0.5^{20} \text{ or}$$

$$2 * [1 + 20 + 190 + 1140 + 4845] * 0.5^{20} \text{ or } 0.0059$$

Do not Reject H_0 at $P = 0.01$

$$\text{Not so Good Answer } Z = \frac{16 - 10}{\sqrt{0.5 * 0.5 * 20}} = \frac{6}{2.23} = 2.68 \quad P = 0.0076$$

Reject H_0 at $P = 0.01$

d) A scientist believes that 25% of adult Busch Bears weigh 11 pounds or less. Is this refuted by the data (explain your answer)?

If this were true the chance to see all 20 adult Busch Bears weigh 12 pounds or more (which is what happened) is $.75^{20} = 0.003 \dots$. This is not very likely. So this belief is refuted by the data.

4. Puntious Pilot the Roman who pioneered kicking egg-shaped air inflated balls is building a linear model for force applied to the ball and the distance the ball travels. The model is $Y = bX + \epsilon$ where X is the force Applied, Y is the distance the ball travels and ϵ is iid error with mean 0 and constant variance σ^2 . **NOTE that the intercept of this model is 0.** Puntious has the following data to use to estimate b

X	1	2	3	4	5	6	7
Y	3	4	5	9	8	20	18

COMPETENCIES = Linear Models, Least Squares Estimates

- a. Write out the design matrix X
 $[1 \ 2 \ 3 \ 4 \ 5 \ 6 \ 7]'$

- b. Estimate b

$$\hat{b} = \frac{\sum XY}{\sum X^2} = \frac{1 \times 3 + 2 \times 4 + 3 \times 5 + 4 \times 9 + 5 \times 8 + 6 \times 20 + 7 \times 18}{1^2 + 2^2 + 3^2 + 4^2 + 5^2 + 6^2 + 7^2} = \frac{348}{140} = 2.49$$

- c. Estimate σ^2

Y	3	4	5	9	8	20	18
$\hat{Y}$	2.49	4.98	7.47	9.96	12.45	14.94	17.43
$(\hat{Y} - Y)^2$	0.24	0.96	6.10	0.92	19.80	25.60	0.34

$$\text{So } \hat{\sigma}^2 = \frac{\sum (\hat{Y} - Y)^2}{n-1} = \frac{0.24 + 0.96 + 6.10 + 0.92 + 19.80 + 25.60 + 0.34}{7-1} = \frac{53.96}{6} = 8.99$$

- d. What is the variance of $\hat{b}$ your estimate for b up above

$$\text{Var}(\hat{b}) = \sum \left(\frac{X}{\sum X^2} \right)^2 \sigma^2 = \frac{1^2 + 2^2 + 3^2 + 4^2 + 5^2 + 6^2 + 7^2}{(140)^2} = \frac{1}{140} 8.99 = 0.064$$

- e. If a new person kicks the ball with force $X = 4$, use the model to give a 95% confidence interval for Y , how far the ball travels.

$$\begin{aligned} 4\hat{b} \pm t_{6,975} \sqrt{\hat{\sigma}^2 (1 + 4^2 \text{var}(\hat{b}))} &= 4\hat{b} \pm t_{6,975} \sqrt{\hat{\sigma}^2 (1 + 4^2 / 140)} = 4 \times 2.49 \pm 2.447 \sqrt{8.99 \times 1.12} \\ &= 9.96 \pm 2.447 \times 3.17 = (2.19, 17.73) \end{aligned}$$

SPRING 2018

1. HIV research scientists are studying the associations of three normally distributed parameters of HIV infection; CD4 count, log-HIV-viral load (LHVL) and P24-Antigen (P24). These are measured on 100 independent subjects and the following sample correlations are observed.

$$r_{\text{CD4,LHVL}} = 0.25$$

$$r_{\text{CD4,P24}} = 0.17$$

$$r_{\text{LHVL,P24}} = 0.35$$

$$\text{NOTE THE INVERSE TRANSFORMAION OF } 0.5 * \ln\left(\frac{1+x}{1-x}\right) \text{ is } \left(\frac{e^{2x}-1}{e^{2x}+1}\right)$$

COMPETENCIES – Statistical properties of correlation coefficients

- a. Use the best test for the null hypothesis that $\rho_{\text{CD4,P24}} = 0$ against the alternative that $\rho_{\text{CD4,P24}} > 0$ at $\alpha = 0.05$

ANSWER test statistic for testing a correlation equals 0 is $\frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$ or $\frac{0.17\sqrt{100-2}}{\sqrt{1-(0.17)^2}} = \frac{1.68}{0.985} = 1.705 > Z_{1-0.05} = 1.65$ so reject H_0

- b. Give the best 95% confidence interval for $r_{\text{CD4,LHVL}}$

Do a 95% confidence interval on Fisher's Z transformation $z_r = 0.5 * \ln\left(\frac{1+r}{1-r}\right)$

$$= z_r = 0.5 * \ln\left(\frac{1+0.25}{1-0.25}\right) = 0.5(0.51) = 0.255. \text{ The variance of } z_r \text{ is}$$

$$\frac{1}{N-3} = \frac{1}{100-3} = \frac{1}{97} \text{ so the 95\% confidence interval for } 0.5 * \ln\left(\frac{1+\rho}{1-\rho}\right) \text{ is}$$

$$0.255 - 1.96\sqrt{\frac{1}{97}} < 0.5 * \ln\left(\frac{1+\rho}{1-\rho}\right) < 0.255 + 1.96\sqrt{\frac{1}{97}}$$

$$0.255 - 1.96(0.101) < 0.5 * \ln\left(\frac{1+\rho}{1-\rho}\right) < 0.255 + 1.96(0.101)$$

$$0.057 < 0.5 * \ln\left(\frac{1+\rho}{1-\rho}\right) < 0.453$$

Converting this back to a 95% CI for ρ is

$$\left(\frac{e^{2(0.057)} - 1}{e^{2(0.057)} + 1} \right) < \rho < \left(\frac{e^{2(0.453)} - 1}{e^{2(0.453)} + 1} \right) \text{ or } \left(\frac{e^{0.114} - 1}{e^{0.114} + 1} \right) < \rho < \left(\frac{e^{0.906} - 1}{e^{0.906} + 1} \right)$$

$$\text{or } \left(\frac{1.12 - 1}{1.12 + 1} \right) < \rho < \left(\frac{2.47 - 1}{2.47 + 1} \right) \text{ or } 0.057 < \rho < 0.424$$

- c. Give a point estimate for the partial correlation of LHV and P24 after adjusting for CD4

$$r_{-LHV, P24 | CD4} = \frac{r_{-LHV, P24} - (r_{-CD4, LHV} \times r_{-CD4, P24})}{\sqrt{1 - (r_{-CD4, LHV})^2} \sqrt{1 - (r_{-CD4, P24})^2}} = \frac{0.35 - (0.25 \times 0.17)}{\sqrt{1 - (0.25)^2} \sqrt{1 - (0.17)^2}}$$

$$= \frac{0.3075}{\sqrt{0.9375} \sqrt{0.9711}} = 0.322$$

NOTE - DEPENDING ON THE TEXTBOOKS USED THERE MAY BE SOME VARIATION IN USE OF T DISTRIBUTIONS INSTEAD OF X AND ON THE FORMULAS FOR THE VARIANCE. ALL WILL BE ACCEPTED.

3. The following data exist on the moisture output from a furnace and the volume of ice forming on the outside tubing over 5 days

Day	1	2	3	4	5
Moisture	5	1	3	2	4
Ice	8	1	4	4	8

Assume the standard linear model $Y = a + bX + e$ where Y is ice, X is moisture and e is iid error with mean 0 and a, b are unknown parameters

COMPETENCIES = Linear Regression

- A. Derive the minimum variance unbiased estimates for a and b

$$\bar{X} = (5 + 1 + 3 + 2 + 4) / 5 = 15/5 = 3$$

$$\bar{Y} = (8 + 1 + 4 + 4 + 8) / 5 = 25/5 = 5$$

$$\hat{b} = \Sigma[(X - \bar{X})(Y - \bar{Y})] / \Sigma(X - \bar{X})^2 \quad (*)$$

Day	1	2	3	4	5
(X - $\bar{X}$)	2	-2	0	-1	1
(Y - $\bar{Y}$)	3	-4	-1	-1	3

$$\text{So } (*) = (2 \times 3 + -2 \times -4 + 0 \times -1 + -1 \times -1 + 1 \times 3) / (2 \times 2 + -2 \times -2 + 0 \times 0 + -1 \times -1 + 1 \times 1)$$

$$= (6 + 8 + 0 + 1 + 3) / (4 + 4 + 0 + 1 + 1)$$

$$= 18 / 10 = 1.8$$

$$a_{\text{hat}} = Y_{\text{bar}} - b_{\text{hat}} X_{\text{bar}} = 5 - 3 \times 1.8 = -0.4$$

B. What is an unbiased estimate for the variance of e?

$$\Sigma (Y - Y_{\text{Hat}})^2 / n-2 \quad (**)$$

Where $Y_{\text{hat}} = a_{\text{hat}} + b_{\text{hat}} X$

So for

$$X=1 \quad Y_{\text{hat}} = -0.4 + 1.8 \times 1 = 1.4$$

$$X=2 \quad Y_{\text{hat}} = -0.4 + 1.8 \times 2 = 3.2$$

$$X=3 \quad Y_{\text{hat}} = -0.4 + 1.8 \times 3 = 5$$

$$X=4 \quad Y_{\text{hat}} = -0.4 + 1.8 \times 4 = 6.8$$

$$X=5 \quad Y_{\text{hat}} = -0.4 + 1.8 \times 9 = 8.6$$

Or

Day	1	2	3	4	5
Y	8	1	4	4	8
Y-Hat	8.6	1.4	5	3.2	6.8

and (**)

$$= [(8 - 8.6)^2 + (1 - 1.4)^2 + (4 - 5)^2 + (4 - 3.2)^2 + (8 - 6.8)^2] / (5-1)$$

$$= [0.36 + 0.16 + 1 + 0.64 + 1.96] / 4$$

$$= 1.20$$

C. It is now proposed that the variance of e is proportional to X in other words $\text{Var}(e) = c * X$ for some constant c. What approach would you use to derive better estimates for a and b. You are being asked to describe the approach well enough to show full understanding, but do not need to work out the mathematical details to actually derive the estimates.

Use weighted least squares

$$[a_{\text{hat}} \ b_{\text{hat}}] = [X' V^{-1} X]^{-1} [X' V^{-1} Y]$$

where

X' is the design matrix

1	1	1	1	1
5	1	3	2	4

Y is the outcome matrix

8	1	4	4	8
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V is a diagonal 5 x 5 matrix with the diagonal elements being

1/5	1	1/3	1/2	1/4
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FALL 2017

2. Dr. Jones is testing a new home delivered education program to increase kindergarten test scores. He will randomly chose 50 independent children to have his program given to them by their parents and then compare the results to 50 other independent children who are not given the program. From normative data, typical education test scored without the intervention have a standard deviation of 15 units and are normally distributed. Higher test scores are better. He will test the null hypothesis that the intervention has not effect on test scores with a two sided $\alpha = 0.05$. Let Δ represent the true amount of units on average that the intervention improves test score in the population.

COMPETENCIES = Power Estimation,

- A. Assume the intervention has no effect on the standard deviation of a test score and give the power to detect a difference of $\Delta=14$ under H_A .

$$(1-\beta) = Z^{-1} \left[\frac{\Delta}{\sigma\sqrt{2/n}} - Z_{1-\alpha/2} \right] = Z^{-1} \left[\frac{14}{15\sqrt{2/50}} - 1.96 \right] = Z^{-1} [4.66 - 1.96] = Z^{-1} [2.70] = 0.9965$$

- B. Now further assume that after the study is done that 21 of the 50 parents who were supposed to give the program to their child did not, so the child had no benefit,

- i. Assuming the 21 parents cannot be identified and all 50 children parents randomized to receive Dr. Jones program must be analyzed together, will power to detect $\Delta = 14$ under H_A be more or less than what is calculated above for Part A.? Explain why

Will have lower power than what is calculated in A. as the actual difference will only be $(29/50) 14 = 8.12$. Furthermore, the standard deviation of the experimental group will increase.

- ii. If the 21 parents can be identified will excluding their children from the 50 and only comparing the 29 who delivered the program to the 50 controls typically improve power over analyzing al 50 together? Explain why or why not

Should improve power as the difference will once again be $\Delta=14$ which

By removing the 21 “contaminated” observations will offset the reduction in sample size to 29 in that arm

✂

4. This data exists on the number of hurricanes that have hit Typhoon Island, a place that lies on the equator and thus has hurricanes throughout the year.

NUMBER OF HURRICANES HITTING TYPHOON ISLAND

Year	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016
Number	2	5	7	1	4	0	9	3	8	5

COMPETENCIES = Central Limit Theorem, Properties of expectation and variance, Poisson distribution

- A. Assume that the number of hurricanes each year is independent with a common standard deviation, but that the intensity can differ randomly from year to year. Then derive a large sample estimate for the expected number of hurricanes in 2017 and the standard deviation of this estimate.

Normal Distribution Central Limit Theorem

$$\text{Mean} = (2+5+7+1+4+0+9+3+8+5) / 10 = 4.4$$

Variance of Estimated Mean

$$= \{[(2^2 + 5^2 + 7^2 + 1^2 + 4^2 + 0^2 + 9^2 + 3^2 + 8^2 + 5^2) - 10 (4.4^2)] / (10-1)\} / 10$$

$$= \{[4 + 25 + 49 + 1 + 16 + 0 + 81 + 9 + 64 + 25] - 193.6\} / 9\} / 10$$

$$= \{(274 - 193.6) / 9\} / 10 = 8.93 / 10 = 0.893$$

$$\text{Std-Err} = \text{SQRT}(0.893) = 0.945$$

- B. Assume that monetary damages from each hurricane is independently distributed including being independent of the number of hurricanes that occur in one year. Use the information on monetary damages from 8 randomly chosen hurricanes below along with your answer to A above to give an estimate for the expected total monetary damages from all hurricanes in 2017

MONETARY DAMAGES FROM 8 RANDOMLY CHOSEN HURRICAINES

Hurricane	1	2	3	4	5	6	7	8
Damages	5	6	4	8	7	6	5	6

For Expected Costs of each Hurricane Based on The Above Table

$$\text{Mean Costs of Damages is } (5+6+4+8+7+6+5+7) / 8 = 5.87$$

Variance of the estimated mean is

$$= \{[(5^2 + 6^2 + 4^2 + 8^2 + 7^2 + 6^2 + 5^2 + 7^2) - 8 (6^2)] / (8-1)\} / 8$$

$$= \{[(25 + 36 + 16 + 64 + 49 + 36 + 25 + 49) - 275.7] / (8-1)\} / 8$$

$$= [(300 - 275.7) / 7] / 8 = 3.47 / 8 = 0.43$$

Expected Expenditures = Expected Number x Expected Cost Per Event

$$= 4.4 \times 5.87 = 25.83$$

- C. Now assume that the number of hurricanes each year is independent with the same intensity over time and derive a large sample estimate for the expected number of hurricanes in 2017 and the standard deviation of this estimate.

Poisson Distribution

$$\text{Mean} = (2+5+7+1+4+0+9+3+8+5) / 2 = 4.4$$

$$\text{Variance} = 4.4 / 10 = 0.44$$

$$\text{Std-Dev} = \text{SQRT}(0.44) = 0.66$$

SPRING 2017

1. THE TOWERS OF THANG-LONG are ceramics that were produced by only two schools in the 14th century, the Borneo-cast and the Malacca-cast. It is very difficult to determine which school had manufactured a given ceramic tower but it is known

Competencies – Bayes Theorem Properties of the Normal Distribution

- height of towers made by the Borneo-cast are distributed $N(25.00, 10.00)$
 - height of towers made by the Malacca-cast are distributed $N(30.00, 5.00)$.
 - also 40% of Thang-Long towers were made by the Borneo cast.
- a. An archeologist discovers a Thang-Long tower that is 27.50 units. Use the above information to estimate the probability that it was made by the Borneo-cast school

$P\{B|B \cup M\} = f\{B\} / [f\{B\} + f\{M\}]$ when M and B are disjoint

$$f(B) = 0.4 \frac{1}{2\pi 5} e^{-\frac{(27.5-25)^2}{2 \times 5^2}} \quad f(M) = 0.6 \frac{1}{2\pi 10} e^{-\frac{(27.5-30)^2}{2 \times 10^2}}$$

$$\frac{0.4 \frac{1}{2\pi 5} e^{-\frac{(27.5-25)^2}{2 \times 5^2}}}{0.4 \frac{1}{2\pi 5} e^{-\frac{(27.5-\mu)^2}{2 \times 5^2}} + 0.6 \frac{1}{2\pi 10} e^{-\frac{(27.5-30)^2}{2 \times 10^2}}} = \frac{e^{-\frac{6.25}{50}}}{e^{-\frac{6.25}{50}} + 0.75 e^{-\frac{6.25}{200}}} = \frac{e^{-0.125}}{e^{-0.125} + 0.75 e^{-0.03125}} =$$

$$\frac{0.882}{0.882 + 0.75(0.969)} = 0.548$$

- b. An archeologist has 5 Thang-Long towers that were all made by the same school the mean height is 27.50 units. Use this to estimate the probability that it was made by the Borneo-cast school.

$$\text{Now } f(B) = 0.4 \frac{1}{2\pi \frac{5}{\sqrt{5}}} e^{-\frac{(27.5-25)^2}{2 \times \left(\frac{5}{\sqrt{5}}\right)^2}} \quad \text{and} \quad f(M) = 0.6 \frac{1}{2\pi \frac{10}{\sqrt{5}}} e^{-\frac{(27.5-30)^2}{2 \times \left(\frac{10}{\sqrt{5}}\right)^2}}$$

$$\frac{0.4 \frac{1}{2\pi \frac{5}{\sqrt{5}}} e^{-\frac{(27.5-25)^2}{2 \times \left(\frac{5}{\sqrt{5}}\right)^2}}}{0.4 \frac{1}{2\pi \frac{5}{\sqrt{5}}} e^{-\frac{(27.5-25)^2}{2 \times \left(\frac{5}{\sqrt{5}}\right)^2}} + 0.6 \frac{1}{2\pi \frac{10}{\sqrt{5}}} e^{-\frac{(27.5-25)^2}{2 \times \left(\frac{10}{\sqrt{5}}\right)^2}}} = \frac{e^{-\frac{6.25}{10}}}{e^{-\frac{6.25}{10}} + 0.75 e^{-\frac{6.25}{40}}} = \frac{e^{-0.625}}{e^{-0.625} + 0.75 e^{-0.156}} = \frac{0.535}{0.535 + 0.75(0.855)} = 0.455$$

- c. What additional information from the measures of the 5 towers in b. could help to produce a better estimate for the probability that it was made by the Borneo-cast school. Only describe the information not why it could be useful

Knowing the standard deviation of the 5 measures would be useful as Malacca-cast towers standard deviation is smaller so a smaller sample standard deviation would put more evidence towards Malacca-cast and a larger standard deviation would support Borneo-cast.

Or better than that, knowing the value for each of the five observations so that an overall likelihood ratio could be constructed as the product of each of the five LR's

4. THE SNAKES OF THE KALAHARI undergo a long migration during the summer months where many of the snakes die. Dr. Lizard-Beth Taylor (LBT) measured pre-migration weight centigrams (cg) and length in centimeters (Cm) and on 100 snakes plus whether or not the snakes survived the migration.

COMPETENCIES – Logistic Regression, Delta Method, Causal Inference

SHE FIRST FITS THIS **LOGISTIC REGRESSION** MODEL FOR SURVIVING MIGRATION THAT USES WEIGHT AS A PREDICTOR

	Estimate	Variance	
Intercept ($\hat{a}$)	-0.7	0.25	$Cov(\hat{a}, \hat{b}) = -0.007$
Weight ($\hat{b}$)	0.2 /cg	0.0004	

A. Snake A is 2 cg heavier than Snake B. Give an OR and 95% confidence interval for the odds ratio of Snake A surviving the migration Vs. Snake B.

Point Estimate is $e^{2(0.2)} = e^{0.4} = 1.49$

For 95% CI do 95% CI for b first which is $0.2 \pm 1.96 * \sqrt{0.0004}$ or $0.2 \pm 1.96 * 0.02$ or (LL=0.16, UL = 0.24)

Then 95% CI is $[e^{2*LL} < OR < e^{2*UL}]$ or $[e^{0.32} < OR < e^{0.48}]$ or $1.37 < OR < 1.62$

B. Give a point estimate and 95% confidence interval for the probability that a 10 Cg snake survives the migration

$$\text{Point Estimate is } \frac{e^{-0.7+10(0.2)}}{1+e^{-0.7+10(0.2)}} = \frac{e^{1.3}}{1+e^{1.3}} = \frac{3.67}{1+3.67} = 0.79$$

For 95% CI, do 95% CI for $\alpha + 10$ (a) first

Point Estimate of $\alpha + 10$ (b) is $-0.7 + 10(0.2) = 1.3$

$$\begin{aligned} \text{StdDev of } \alpha + 10 (\beta) &= \sqrt{\text{Var}(\alpha) + 10^2 (\text{Var}(\beta)) + 2 * 10 (\text{Cov}(\alpha, \beta))} \\ &= \sqrt{0.25 + 100 * 0.0004 - 2 * 10 * 0.007} \\ &= \sqrt{0.15} = 0.39 \end{aligned}$$

So 95% CI for $\alpha + 10$ (β) is $1.3 \pm 1.96 * 0.39$ or (LL=0.53 to UL=2.06)

$$\begin{aligned} \frac{e^{LL}}{1+e^{LL}} < S < \frac{e^{UL}}{1+e^{UL}} \text{ is a 95\% confidence interval} \\ \frac{e^{0.53}}{1+e^{0.53}} < S < \frac{e^{2.06}}{1+e^{2.06}} \text{ or } \frac{1.70}{1+1.70} < S < \frac{7.84}{1+7.84} \text{ or } 0.63 < S < 0.88 \end{aligned}$$

SHE THEN FITS TWO MORE MODELS FOR SURVIVING THE MIGRATION WITH [LENGTH] AND [LENGTH + WEIGHT] AS PREDICTORS

	Estimate	Variance			Estimate	Variance
Intercept	1.3	0.32		Intercept	-0.4	0.50
Length	-0.05	0.004		Length	-0.6	0.003
				Weight	0.3	0.0003

C. Use the information on all Three models fitted to determine the best answer for if there is a causal association between length and probability to survive

ANSWER IS NO

From the model with only length, the Z-Value for weight is $-0.05 / \sqrt{0.004} = -0.05 / 0.2 = -0.25$ which has a two sided P-Value of 0.80

NOTE – The adjusted model using Weight and Length is not a good model to use. Since we know that longer lizards have bigger body volume and thus are heavier. We know that heavier weight is associated with longer survival. It stands to reason that if two lizards have the same weight that the longer lizard is “skinnier” and hence if it is skinnier lizards are more likely to die then length should have a negative coefficient after adjusting for weight.

FALL 2016

1. People want to compare two plantations for pesticide levels in bananas. They have tested the levels of 6 randomly chosen bunches of bananas each from these two plantations. From each bunch, two randomly chosen bananas are tested. The following pesticide level results are observed for the two bananas within each Bunch

COMPETENCIES = Linear Models / Design of Experiments (Nested Design)

PLANTATION A - Bunch-1 20, 24 Bunch-2 30, 24 Bunch-3 19, 19 Bunch-4 12, 18 Bunch-5 27, 21 Bunch-6 15, 23

PLANTATION B - Bunch-1 20, 14 Bunch-2 20, 24 Bunch-3 16, 14 Bunch-4 17, 13 Bunch-5 17, 15 Bunch-6 19, 17

You can assume the observations are independently normally distributed with equal variance

MODEL $Y_{ijk} = \mu + b_i I_{\{\text{From Plantation B Effect}\}} + \gamma_j(\text{Plantation}) + \epsilon_{ijk}$

Where $i = A, B$

$J = 1, \dots, 6$

$k = 1, 2$

NOTE No one should consider the repeated bananas in the same bunch as independent without including the γ_j Bunch Effect

From the data we get $\bar{Y}_{A1} = 22, \bar{Y}_{A2} = 27, \bar{Y}_{A3} = 19, \bar{Y}_{A4} = 15, \bar{Y}_{A5} = 24, \bar{Y}_{A6} = 19, \bar{Y}_{A..} = 21$
 $\bar{Y}_{B1} = 17, \bar{Y}_{B2} = 22, \bar{Y}_{B3} = 15, \bar{Y}_{B4} = 14, \bar{Y}_{B5} = 16, \bar{Y}_{B6} = 18, \bar{Y}_{B..} = 17$

The pooled variance of $\bar{Y}_{B..} Y_{i..}$ is

$$= \frac{(22-21)^2 + (27-21)^2 + (19-21)^2 + (15-21)^2 + (24-21)^2 + (19-21)^2 + (17-17)^2 + (22-17)^2 + (15-17)^2 + (14-17)^2 + (18-17)^2 + (17-17)^2}{(6-1) + (6-1)}$$

$$= \frac{1+36+4+36+9+4+0+25+4+9+1+0}{10} = 12.7$$

So $\hat{\sigma} = \sqrt{12.7} = 3.56$ with 10 degrees of freedom

A. Give the best 95% confidence interval for the mean level of pesticides in Bananas from Plantation A

$$\bar{Y}_{A..} \pm t_{10,0.975}(\hat{\sigma} / \sqrt{6}) \text{ or } 21 \pm 2.23 \times (3.56 / 2.45) \text{ or } (17.67, 24.24)$$

NOTE – Some people may base the variance only on $\bar{Y}_{A1}, \bar{Y}_{A2}, \bar{Y}_{A3}, \bar{Y}_{A4}, \bar{Y}_{A5}, \bar{Y}_{A6}$ about $\bar{Y}_{A..}$ which is 17.4 or std-dev = 4.17 in which case they should also use $t_{5,0.975} = 2.57$

- B. Give the best statistical test of whether mean pesticide levels differ in Plantation A and B at a two sided $\alpha=0.05$

$$t_{10df} = \frac{(\bar{Y}_{B..} - \bar{Y}_{A..})}{\hat{\sigma} \sqrt{1/6 + 1/6}} = \frac{(21-17)}{3.56 \sqrt{1/3}} = \frac{4}{2.05} = 1.95 \quad p > 0.05 \text{ for } 10 \text{ d.f.}$$

4. Three types of Fish live in Rutgers pond; Busch-fish, Livingston-fish and Douglas-fish. Drs. HD and DH each recorded fish caught in Rutgers pond from randomly chosen samples.

- Out of a random sample of 120 fish recorded by Dr. HD, 40 were Douglas, 40 were Livingston and 40 were Busch.
- Dr. DH recorded results from a different independent random sample. But he did not record the number of Busch fish Thus out of 90 fish that were either Douglas or Livingston, Dr. HD recorded 60 Livingston and 30 Douglas

Assume that the fish recorded by Drs. HD and DH both random, independent of each other and obtained under the same probabilities to be a Douglas, Livingston or a Busch.

COMPETENCIES = Binomial Distributions, Bayes Theorem and related statistical properties

- A. Use this data from Drs. HD and DH to find the best point estimates for the probability that the species of a fish caught at Rutgers pond were i) Douglas, ii) Livingston and iii) Busch

The only sample that can be used to estimate probability a fish is a Busch is from Dr. HD which is $40/120 = 1/3$

The probability to be a Douglas is determined by both samples

$$\begin{aligned} & P\{\text{Not-Busch}\} \quad * \quad P\{\text{Douglas} \mid \text{Not-Busch}\} \\ & \quad 80 / 120 \quad * \quad (40 + 30) / (80 + 90) \\ & \quad \text{From HD} \quad \quad \quad \text{from DH \& HD} \\ = & \quad 2/3 \quad * \quad 70/170 \quad = 14/51 \end{aligned}$$

The probability to be a Livingston is determined by both samples

$$\begin{aligned} & P\{\text{Not-Busch}\} \quad * \quad P\{\text{Livingston} \mid \text{Not-Busch}\} \\ & \quad 80 / 120 \quad * \quad (40 + 60) / (80 + 90) \end{aligned}$$

$$= \frac{2}{3} \times \frac{100}{170} = \frac{20}{51}$$

- B. Give a large sample 95% confidence interval for the probability that the species of a fish caught at Rutgers Pond is a Busch

$$\text{Use } \text{Var}(A * B) = \text{Var}(A) * [E[B]]^2 + \text{Var}(B) * [E[A]]^2$$

Where

A is probability to not be Busch

B is probability to be Douglas | Not-Busch

$$E[A] = 80/120 \quad \text{Var}[A] = [80/120 * 40/120]/120 = 2/1,080$$

$$E[B] = 7/17 \quad \text{Var}[B] = [7/17 * 10/17]/170 = 70/49,130 = 7/4,913$$

$$\text{So } \text{Var}(A*B) = [80/120]^2 * 7/4,913 + [7/17]^2 * 2/1,080 = 0.000947$$

$$\text{Or Std-Dev}(A*B) = (0.000947)^{0.5} = 0.031$$

$$\text{Thus 95\% CI is } 14/51 \pm 1.96 * 0.031 = (0.2745 - 0.0608, 0.2745 + 0.0608) \\ (0.2137, 0.3353)$$

ALTERNATE METHOD USING DELTA METHOD ON LOG TRANSFORMED DATA

$$\text{Var}[\text{Log}(A * B)] = \text{Var}(\text{Log}(A)) + \text{Var}(\text{Log}(B))$$

$$\text{Var}[\text{Log}(A)] = \text{Var}(A)/[E[A]]^2 = 2/1,080 / [80/120]^2 = 9 / 2,160$$

$$\text{Var}[\text{Log}(B)] = \text{Var}(B)/[E[B]]^2 = 7/4,913 / [7/17]^2 = 289 / 34,391$$

$$\text{So } \text{Var}[\text{Log}(A * B)] = 9/2,160 + 289/34,391 = 0.01257$$

$$\text{Std-Dev}[\text{Log}(A * B)] = (0.01257)^{0.5} = 0.112$$

$$\text{So 95\% CI } \log(A*B) \text{ is } \log(14/51) \pm 1.96 * 0.112 = -1.29 \pm 0.22 = (-1.51, -1.07)$$

Exponenting back is $e^{-1.51}$, $e^{-1.07}$ or (0.22,

SPRING 2016

1. An investigator reads a journal article on Total-cholesterol, HDL-cholesterol and LDL-cholesterol from a study of 100 patients where Total-cholesterol = HDL + LDL cholesterol. Assume that for HDL, LDL and Total cholesterol the distributions are iid multivariate normal. The article reports for:

HDL: sample-mean = 145 and p-value for the population mean to be 150 is 0.10

LDL: sample-mean = 48 and p-value for the population mean to be 50 is 0.05

TOTAL: sample-mean = 193 and p-value for the population mean to be 200 is 0.05

All P-values are TWO-SIDED

COMPETENCY = Basic properties of normal distribution and means / covariances

He must derive standard deviations and covariances for HDL and LDL from this information

- A. Work backwards from the p-values for the populations means equaling the specified values based on the observed sample means to obtain the standard deviations for both HDL and LDL in the observations?

For HDL

$$\frac{|150-145|}{\sigma/\sqrt{100}} = Z_{1-0.10/2} \text{ or } \frac{5}{\sigma/\sqrt{100}} = 1.65 \text{ or } \frac{5}{1.65/10} = \sigma \text{ or } \sigma = \frac{50}{1.65} = 30.3$$

For LDL

$$\frac{|50-48|}{\sigma/\sqrt{100}} = Z_{1-0.05/2} \text{ or } \frac{2}{\sigma/\sqrt{100}} = 1.96 \text{ or } \frac{2}{1.96/10} = \sigma \text{ or } \sigma = \frac{20}{1.96} = 10.2$$

- B. What is the covariance of HDL and LDL cholesterol in a single observation?

Since $\text{HDL} + \text{LDL} = \text{TOTAL}$

$$\text{Var}(\text{TOTAL}) = \text{Var}(\text{HDL}) + \text{Var}(\text{LDL}) + 2*\text{Cov}(\text{HDL}, \text{LDL})$$

$$\begin{aligned} \text{But from A we just saw that } \text{Var}(\text{HDL}) &= (30.3)^2 = 918 \\ \text{Var}(\text{LDL}) &= (10.2)^2 = 104 \end{aligned}$$

To get the Variance of the TOTAL

$$\frac{|200-193|}{\sigma/\sqrt{100}} = Z_{1-0.05/2} \text{ or } \frac{7}{\sigma/\sqrt{100}} = 1.96 \text{ or } \frac{7}{1.96/10} = \sigma \text{ or } \sigma = \frac{70}{1.96} = 35.7$$

$$\text{or Var(HDL)} = (35.7)^2 = 1,274$$

$$\begin{aligned}\text{So } 1,274 &= 918 + 104 + 2 * \text{Cov(HDL, HDL)} \\ 1,274 - (918 + 104) &= 2 * \text{Cov(HDL, HDL)} \\ 252 &= 2 * \text{Cov(HDL, HDL)} \\ \text{Cov(HDL, HDL)} &= 252/2 = 126\end{aligned}$$

2. Dawn-Holly plants have male and female genders. The time for a single female Dawn-Holly plant grown alone to bloom from birth (i.e. emergence from the ground) in weeks is distributed exponential with $\lambda=10$; $f(t) = (1/10) e^{-t/10}$. The time for a single male Dawn-Holly grown alone to bloom is exponential with parameter $\lambda=15$; $(1/15) e^{-t/15}$. However, if a female plant is grown paired with (i.e. within 50 feet of) a male plant the time to blooming for both plants is independent except that once the male plant blooms, the pollen released by the male triggers the female plant to immediately bloom if the female plant did not bloom first. Thus when a male and female Dolly-Holly are paired together, the time for the female plant is $\min(Y, Z)$ where $Y \sim \exp(10)$ and $Z \sim \exp(15)$.

COMPETNECY = Properties of exponential and normal approximation

- A. Compared to growing a female Dolly-Holly alone, by how much time on average does pairing the female plant with a male Dolly-Holly plant reduce the expected time to bloom for a female Dolly-Holly?

The mean of the time to blooming for a female plant that is not paired is $1/(1/10) = 10$ weeks.

When paired with the main plant, the CDF of time to bloom for the female plant is $e^{-t/10} \times e^{-t/15} = e^{-5t/30} = e^{-t/6}$. So the mean time to bloom is now 6 weeks.

This means that adding the male plant has reduced the time to bloom on average by 4 weeks = 10 – 6 weeks

- B. If $N=100$ independent pairs of female and male Dolly-Holly plants are grown, Let X be then number of pairs where the male Dolly-Holly bloomed first and stimulated the female plant to bloom. What is the probability that $X \geq 50$.

For each pair, the probability the male blooms first is $4/10$ derived as follows. The probability the male blooms first is

$$\begin{aligned}\int_0^{\infty} f(\text{male blooms at } x) (\text{Pr ob female Not Bloomed By } X) &= \\ \int_0^{\infty} (1/15)e^{-t/10} e^{-t/15} dt &= \int_0^{\infty} (1/15)e^{-(t/10+t/15)} dt = \int_0^{\infty} (1/15)e^{-(t/6)} dt = \left[-6/15 e^{-(t/6)} \right]_0^{\infty} \\ &= 6/15 = 0.4\end{aligned}$$

So using the normal approximation to the binomial, the probability to observe 50 or more pairs (0.5 of the 100 pairs) in which the male blooms first is 1 minus the CDF of Z where

$$Z = \frac{0.5 - 0.4}{\sqrt{(0.4)(1-0.4)/100}} = \frac{0.1}{0.0489} = 2.04 \text{ and } Z^{-1}(2.04) = 0.9793 \text{ so the answer is}$$

$$1 - 0.9793 = 0.02$$

FALL 2015

1. Dr. Jones obtained least squares linear regression estimates for parameters of this linear model with an interaction term $Y = a + b_1 X_1 + b_2 X_2 + b_{12} X_1 X_2 + \epsilon$ using 94 adults; where Y is hematocrit, X_1 is sex (0=female, 1=male) X_2 is **[age in years – 30]**; a , b_1 , b_2 , and b_{12} are unknown constants; and ϵ is normally distributed error with mean 0 and unknown variance σ^2 and obtained the following estimates. You can assume that all of these previous assumptions are valid when you answer the questions.

PARAMETER	ESTIMATE	STD-ERROR
a	40	1.1
b_1	-0.1	0.025
b_2	-2	0.5
b_{12}	-0.5	0.025

and $\hat{\sigma}^2 = 10$

- A. Give a 95% confidence interval for σ^2 or if you cannot answer the question from the information given explain what additional information you would need to answer the question

COMPETENCY = LINEAR MODELS

$(\hat{\sigma}^2 / \sigma^2) \times 90$ has a chisquare distribution with $94-4 = 90$ d.f.

So $(\chi^2_{(94,0.975)} / 90) \hat{\sigma}^2 < \sigma^2 < (\chi^2_{(94,0.025)} / 90) \hat{\sigma}^2$ is a balanced 95% confidence

interval for σ^2 where $\chi^2_{(94,0.975)} = 65.647 / 90 = 0.73$ and

$\chi^2_{(94,0.025)} / 90 = 118.136 / 90 = 1.31$

thus $0.73 * 10 < \sigma^2 < 1.31 * 10$

$7.3 < \sigma^2 < 13.1$ is a balanced 95% CI for σ^2

- B. Give a 95% confidence interval for the mean hemoglobin of 30 year old women or if you cannot answer the question from the information given explain what additional information you would need to answer the question

Start By Getting the Expected Value First:

For a 30 years old woman, $X_1 = X_2 = 0$ so $E[Y] = a + b_1 X_1 + b_2 X_2 + b_{12} X_1 X_2$

$E[Y] = a + b_1 0 + b_2 0 + b_{12} 0 = a$

And Now you can see that the 95% CI for a is the 95% confidence interval for $E[Y]$ of a 30 year old woman Since $df = 90$ which is > 60 you can use Z statistic and

$\hat{a} - 1.96(std - err(\hat{a})) < a < \hat{a} + 1.96(std - err(\hat{a}))$

$40 - 1.96(1.1) < a < 40 + 1.96(1.1)$ or $40 - 2.16 < a < 40 + 2.16$ or

$(37.84, 42.16)$ is the 95% confidence interval for the expected hematocrit of a 42 year old woman

4. Dr. Lee developed The Dooku-Test for infection by the Dooku virus.
- I. He tests 1,000 persons who truly have the Dooku virus The Dooku-Test
 - correctly identifies that 990 have the Dooku virus
 - falsely identifies that 10 do not have the Dooku virus
 - II. He then tests 1,000 persons who don't have the Dooku virus The Dooku-Test
 - correctly identifies that 990 do not have the Dooku virus
 - falsely identifies that 10 do have the Dooku virus

Remember that the numbers in I and II are sample estimates

Suppose in the United States that exactly 1/1000 of the persons (0.001) are infected with the Dooku virus which means 0.999 are not. You should take this number to be known and not an estimate

COMPETENCIES = Applied Probability, Normal Approximation to Binomial, Bayes Theorem

- A. Give the best point estimates for
- a. probability that a US person is both infected with the Dooku virus and would test positive for Dooku by the Dooku-Test.

ANSWER

$$P\{\text{infected}\} \times P\{\text{test positive} \mid \text{infected}\} \\ 0.001 \quad * \quad 0.99 \quad \quad \quad = 0.00099$$

- b. Give the best point estimated for the probability that a US person is both not infected with the Dooku virus and would test positive for Dooku by the Dooku Test.

ANSWER

$$P\{\text{Not infected}\} \times P\{\text{test positive} \mid \text{not infected}\} \\ (1-0.001) \quad * \quad 0.01 \quad \quad \quad = 0.0099$$

- B. Give a large sample 95% confidence interval for the point estimate in A-a. up above

It is known that 0.001 are infected so with Dooku this is a constant. The 0.99 probability for a person infected with Dooku to test positive by the Dooku test is a point estimate. The 95% large

sample CI is $0.99 \pm 1.96 \sqrt{\frac{0.01 * 0.99}{1,000}}$ or 0.99 ± 0.003 or $0.987 - 0.993$.

Since the probability to be infected with the Dooku virus was given as a constant, we just use it to multiply these limits

Thus $[0.001 \times 0.987]$ to $[0.001 \times 0.993]$ or **0.000987 – 0.000993** is a 95% confidence interval for a US person to be infected with Dooku and to test positive

- C. If a US person tests positive for the Dooku virus, give the best point estimate for the probability they are really infected with that virus.

$$\text{From Bayes Theorem } P\{A \mid (A \cup B)\} = \frac{P\{A\}}{P\{A\} + P\{B\} - P\{A \cap B\}}$$

$$P\{A\} + P\{B\} - P\{A \cap B\} \quad \text{where}$$

A = Person is Truly positive and tests positive $P\{A\} = 0.0099$

B = Person is Truly negative and tests positive $P\{B\} = 0.00099$

$P\{A \cap B\} = 0$ since a person can't be both truly positive and Truly negative at the same time

$$\text{So } P\{A \mid (A \cup B)\} = \frac{0.00099}{0.00099 + 0.099} = \frac{1}{11} = 0.0909$$

SPRING 2015

4. A researcher wants to compare the effectiveness of three anti-wrinkle creams as used for different age groups. The response variable is a numerical measure on a ratio scale of the reduction of wrinkles. 6 doctors participate in the study. 2 doctors are assigned at random to each cream. Each doctor has 4 participating patients aged 40-45, 4 participating patients aged 50-55, and 4 participating patients aged 60-65. The researcher views the doctors as a random sample of doctors, and the patients as a random sample of their patients within those age groups. The data layout is as follows.

	Cream 1		Cream 2		Cream 3	
	Doctor 1	Doctor 2	Doctor 3	Doctor 4	Doctor 5	Doctor 6
40-45	4	4	4	4	4	4
50-55	4	4	4	4	4	4
60-65	4	4	4	4	4	4

Write out the ANOVA table showing source of variability, degrees of freedom, Expected Mean Squares, and show which F ratios to take.

COMPETENCIES

Linear Models

Design of Experiments

BEST WAY TO ANSWER THIS IS TO HAVE A BOOK THAT HAS A LAYOUT FOR THIS TYPE OF ANOVA DESIGN IF THEY ARE STILL PRINTED

NOTE – Doctor is nested within cream

i enumerates age group

J enumerates cream

k enumerates doctor nested within cream

l enumerates patient

$3 \times 2 \times 3 \times 4 = 72$ subjects

Model is $Y_{ijkl} = u + a_i + b_j + c_{k(j)} + \epsilon_{ijkl}$ with $\epsilon_{ijkl} \text{ iid } N(0, \sigma_e^2)$

Where u has 1 degree of freedom

$\sum a_i = 0$ so $3-1 = 2$ degrees of freedom, but as random effects, $a_i \text{ iid } N(0, \sigma_a^2)$,

$\sum b_j = 0$ so $3-1 = 2$ degrees of freedom, but as random effects, $b_j \text{ iid } N(0, \sigma_b^2)$,

Within each $j = 1, 2, 3$ $\sum c_{k(j)} = 0$ so $3 \times (2-1) = 3$ degrees of freedom
but as random effects, $c_{k(j)} \sim \text{iid } N(0, \sigma_c^2)$,

So model has $1 + 2 + 2 + 3 = 8$ degrees of freedom. Residual d.f. = $72 - 8 = 64$

ANOVA TABLE

SOURCE	SUM-OF-SQUARES	D.F	MS	EMS
AGE	$3 \times 2 \times 4 \sum_{i=1}^3 (\bar{Y}_{i..} - \bar{Y}_{...})^2$	2	MS_a	$\sigma_e^2 + 24 \sigma_a^2$
CREAM	$3 \times 2 \times 4 \sum_{j=1}^3 (\bar{Y}_{.j.} - \bar{Y}_{...})^2$	2	MS_b	$\sigma_e^2 + 4 \sigma_b^2 + 24 \sigma_c^2$
DOCTOR	$3 \times 4 \sum_{j=1}^3 \sum_{k=1}^2 (\bar{Y}_{.jk} - \bar{Y}_{.j.})^2$	3	MS_c	$\sigma_e^2 + 4 \sigma_b^2$
RESIDUAL	$\sum_{i=1}^3 \sum_{j=1}^3 \sum_{k=1}^2 \sum_{l=1}^4 (Y_{ijkl} - \bar{Y}_{i..} - \bar{Y}_{.jk} + \bar{Y}_{.j.})^2$	64	MS_e	σ_e^2
TOTAL	$\sum_{i=1}^3 \sum_{j=1}^3 \sum_{k=1}^2 \sum_{l=1}^4 (Y_{ijkl} - \bar{Y}_{....})^2$	71		

F – Ratios to Take

- MS_a / MS_e to Test Age Effect
- MS_b / MS_e to Test Cream Effect
- MS_c / MS_e to Test Doctor Effect

FALL 2014

1. Below are data on numbers of AIDS Cases, number of AIDS Cases caused by PCP and KS for North American cities. (Note AIDS Being Caused by PCP and by KS is mutually exclusive).

CITY	AIDS CASES		
	TOTAL	CAUSED BY PCP	CAUSED BY KS
Des Moines	200	30	30
El Paso	200	20	50

Let p_D be the portion of all AIDS cases in Des Moines caused by PCP

p_E be the portion of all AIDS cases in El Paso caused by PCP

k_D be the portion of all AIDS cases in Des Moines caused by KS

k_E be the portion of all AIDS cases in El Paso caused by KS

COMPETENCIES

Two BY Two Table Word Problem – Question a

Conditional Chisquare Approximation Word Problem – Question b

- a. Dr. LP believes that $p_E = p_D$. Give a large sample test of this alternative hypothesis against the appropriate null at $\alpha=0.05$

TWO BY TWO TABLE (WITHOUT CONTINUITY CORRECTION)

	AIDS - PCP+	AIDS - PCP+	TOTAL
Des Moines	30	170	200
El Paso	20	180	200
TOTAL	50	350	400

$$\frac{(180 * 30 - 170 * 20)^2}{50 * 350 * 200 * 200} = 2.28 \quad P > 0.05 \text{ two sided } H_0$$

- b. Dr. PL believes $k_D = 2p_D$. Give a large sample test of this alternative hypothesis against the appropriate null at $\alpha=0.05$

CONDITIONAL TWO BY TWO TABLE (WITHOUT CONTINUITY CORRECTION)

	AIDS - PCP	AIDS - KS	TOTAL
Des Moines	30	30	60

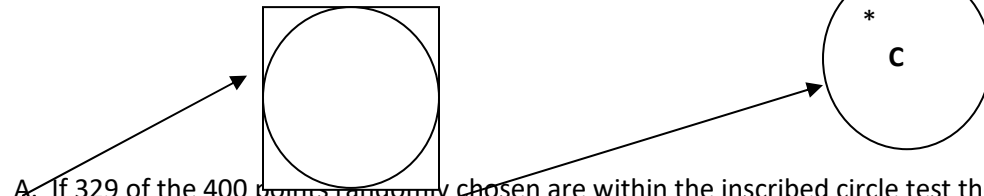
Conditional on 30 PCP + KS cases = 60 total in Des Moines, under the null hypothesis the expected number of PCP and KS cases are 20 and 40 each

$$\sum \frac{(O - E)^2}{E} = \frac{(30 - 20)^2}{20} + \frac{(30 - 40)^2}{40} = 5 + 2.5 = 7.5$$

Compare this to 3.84 the upper (1-0.05) level of a χ^2_{1df} and reject H_0

SPRING 2014

2. 400 points are independently and randomly chosen on a square.



A. If 329 of the 400 points randomly chosen are within the inscribed circle test the null hypothesis of the probability to fall within the circle is random versus the alternative that it is greater than by chance alone relative to area of the circle at $\alpha = 0.05$.

B. Now consider only the 329 independent points inside the inscribed circle. Suppose the radius of this circle is 1 and the sample mean distance of a point from the center of the circle (C) is $\bar{X} = 0.65$ with a variance of $S^2 = 0.056$ (you can use this variance to estimate the variance for the population). Give a large sample test of the two sided null hypothesis on whether or not the points are closer to the center of the circle than they should be by chance alone if the points were randomly distributed inside the circle.

COMPETENCIES

Single Proportion Hypothesis Normal Approximation Word Problem - Question A
Using Calculus to Find Expected Value and Normal Approximation – Question B

2. ANSWER TO A

i). If r is the radius of the inscribed circle, then $2r$ is the height and width of the square. The area of the inscribed circle is πr^2 and the area of the square is $4r^2$. So by chance alone, the probability to be inside the circle is $\pi r^2 / 4r^2 = \pi/4 = 0.785$. This means if 400 points are independently and randomly chosen, and expected 0.785 will be inside the circle and the variance of this observed portion is $0.785 * 0.215 / 400$.

ii) In 400 tries 329 or 0.8225 of the points were inside the circle.

iii) Assume the Null Hypothesis and do a Z test

$$Z = \frac{0.8225 - 0.785}{\sqrt{0.785 * 0.215 / 400}} = \frac{0.0375}{0.0205} = 1.83$$

Since $1.83 > 1.65 = Z_{0.95}$ reject H_0 for H_a that there is a greater than expected (by relative area) chance for a point to be inside the inscribed circle

ANSWER TO B

- i) The expected distance of a point from the Center of the Circle is:

$$\frac{\int_0^1 x(2\pi x)dx}{\pi} = \frac{\int_0^1 2\pi x^2 dx}{\pi} = \frac{\left|_0^1 (2/3)\pi x^3\right.}{\pi} = 0.667$$

- ii) Using The Z-Test, $Z = \frac{0.65 - 0.667}{\sqrt{0.056/329}} = \frac{-0.0167}{0.0130} = -1.23$

- iii) Since $|-1.23| < 1.96$ do not reject H_0

NOTE - Some students may figure out by using calculus to find $E[x^2]$ that if the distribution of points is random, then the **variance** of the distance from the center of the circle is $\frac{1}{2} - 4/9 = 0.0555555.....$ so the “standard deviation” being 0.056 is likely a misprint. If so and a student used 0.056 as the variance rather than the standard deviation they would get And in fact would probably get most to full credit even if 0.056 did turn out to be the standard deviation, but the student showed their calculations as to why they thought it was the variance.

FALL 2013

1. Assume that birthdays are iid distributed across 365 days of the year (exclude Feb 29th on leap year). We have two groups of people

G1 – Has 10 Persons

G2 – Has 10 different persons

Applications of Probability

- A. What is the probability that no two persons in G1 were born on the same day of the year?

$$364/365 \times 363/365 \times \dots \times 356/365 = 0.883$$

- B. Assume that no two persons in G1 have the same birthdate, then what is the probability that none of the persons in G2 were born on the same day as a person in G1?

$$(355/365)^{10} = 0.757$$

- C. Assume again that no two persons in G1 have the same birthdate, what is the probability that none of the 20 persons in G1 and G2 combined have the same birthdate?

$$355/365 \times 354/365 \times \dots \times 346/365 = 0.667$$

SPRING 2013

3. A Statistics Professor believes that adding videos of statisticians consulting on applied problems to her standard instructional material will help students solve problems. She designs an experiment to find evidence to support this hypothesis. A total of 167 students are given a test prior to receiving her class (the “pretest”) to see how they approach applied statistics problems. The pretest is graded Pass or Fail. The students are then randomly divided into two groups: The Experimental Group with 106 students, and the Control Group with 61 students. Over the next 2 weeks, the Experimental group are shown videos plus her standard instructional material, while the Control group receives only her standard instructional material. After this is done, all students take the same Pass/Fail test again on solving applied statistics problems (the “post-test”). You can assume that all students are independent. The following are the combined Pretest Posttest outcomes:

Group	Pretest	Posttest	Number of subjects
Experimental	Fail	Fail	37
Experimental	Fail	Pass	29
Experimental	Pass	Fail	11
Experimental	Pass	Pass	29
Control	Fail	Fail	26
Control	Fail	Pass	9
Control	Pass	Fail	7
Control	Pass	Pass	19

- Using only the data for the experimental group is there statistically significant evidence of an improvement between Pretest and Posttest performance. What is the Null Hypothesis; the Alternative Hypothesis, and show the details for the test used.
- Using only the data for the control group is there statistically significant evidence of an improvement between the pretest and posttest?
- Now combine the analyses for the experimental and control groups to see if there is statistically significant evidence that adding the video to the instruction was helpful.

COMPETENCY

McNemars Discordant Pairs

Two by Two Table Word Problem

ANSWER

H_0 : Probability to Pass post video is the same as the probability to pass pre-video

H_A : Probability to Pass post video is higher than probability to pass pre-video

- a)_ Two by Two Table of Matched Responses in Experimental Group

Pretest Result	Posttest Result	
	Fail	Pass

Fail	37	29
Pass	11	29

Focus on the discordant pairs $11 + 29 = 40$. If H_0 is true, conditional on 40 discordant pairs, the expected number who fail pre and pass post = number who pass pretest and fail post test = $40 / 2 = 20$. If H_A true, the number who pass post and fail pre is larger.

Test Statistic is $\sum \frac{(E-O)^2}{E} = \frac{(29-20)^2}{20} + \frac{(11-20)^2}{20} = 8.1$ Compare to $\chi^2_1(1-0.05) = 3.84$. $P < 0.05$

b)_ Two by Two Table of Matched Responses in Control Group

H_0 : Probability to Pass post video is the same as the probability to pass prevideo

H_A : Probability to Pass post video is higher than probability to pass prevideo

Pretest Result	Posttest Result	
	Fail	Pass
Fail	26	9
Pass	7	19

Focus on the discordant pairs $7 + 9 = 16$. If H_0 is true, conditional on 8 discordant pairs, the expected number who fail pre and pass post = number who pass pretest and fail posttest = $(7+9)/2 = 8$. If H_A true, the number who pass post and fail pre is larger.

Test Statistic is $\sum \frac{(E-O)^2}{E} = \frac{(9-8)^2}{8} + \frac{(7-8)^2}{8} = 0.25$ Compare to $\chi^2_1(1-0.05) = 3.84$. $P > 0.05$

c) There are some potential problems with any answer to this that would be easy enough to complete on the exam. I believe this answer which tests if the matched OR for discordant pairs was the same for each treatment arm would get full credit. Comparative two by two table of discordant results in experimental versus control arm.

H_0 : Ratio of Probability to Pass-post / Pass-pre video in the discordant pairs is the same for both Experimental and Control arms

H_0 : Ratio of Probability to Pass-post / Pass-pre video in the discordant pairs is greater for Experimental arm

Pretest Result	Posttest Result	
	Fail-Pre Pass-Post	Pas-Pre Fail-Post
Experimental	29	11
Control	9	7

38

18

56

Test of Independence $\frac{(29*7 - 9*11)^2 56}{38*18*40*16} = 1.38$ Compare to $\chi^2_1(1-0.05) = 3.84$. $P > 0.05$

FALL 2012

2. Researchers seek to estimate changes in the total number of polar bears in particular regions. Polar bears hunt on floes of ice in the seas off the coast of the “Arctic ring of life” where seals are abundant. One region is by the coast of the Southern Beaufort Sea. This is an area, 1000 kilometers in length by 200 kilometers in width (land on coast plus adjacent sea). The 200,000 square kilometer region was divided into a grid of 8000 subsquares, each 5 kilometers in length by 5 kilometers in width. In 1986 a random sample of 400 subsquares were flown over, and a total of 90 polar bears were observed. In 2006 an independent random sample of 400 subsquares were flown over, and a total of 75 polar bears were observed. You can assume that the polar bears are distributed completely at random over the region.

- Estimate the total number of polar bears in the region in 1986, and in 2006.
- For each estimate, give a standard error of the estimate
- Test the null hypothesis at the 0.10 level of significance of no change (between 1986 and 2006) in number of polar bears in the region, versus the research hypothesis that there has been a decrease in the number of polar bears.

COMPETENCIES

Sampling (possibly including finite populations)

Poisson Approximation

Conditional Probability

ANSWER

To Part a)

400 subsquares is $1/20^{\text{th}}$ of the total 8000 subsquares so

In 1986, 90 is $1/20^{\text{th}}$ of expected total or expected total = $20 \times 90 = 1800$

In 2006, 75 is $1/20^{\text{th}}$ of expected total or expected total = $20 \times 75 = 1500$

To Part b) based on Poisson approximation with λ being the expected number of polar bears in 400 subsquares

THIS ANSWER SHOULD GET MOST IF NOT ALL CREDIT

In 1986 $\hat{\lambda} = 90$ so standard error of $\hat{\lambda} \approx \sqrt{\hat{\lambda}} = \sqrt{90} = 9.49$ So the
Standard error of the Estimated Total is $20 \times 9.49 = 189.8$

In 2006 $\hat{\lambda} = 75$ so standard error of $\hat{\lambda} \approx \sqrt{\hat{\lambda}} = \sqrt{75} = 8.66$ So the
Standard error of the Estimated Total is $20 \times 8.66 = 173.2$

BUT ALSO NOTE THE “MENSA” ANSWER – The grader could decide that in 1986 and 2006 the answers are known to be 90 and 70 in the 400 subsquares and it is only the 7,600 other subsquares where the additional numbers are being estimated with the ratio $7,600 / 400 = 19$. In that case

In 1986 the Standard error of the Estimated Total is $19 * 9.49 = 180.3$

In 2006 the Standard error of the Estimated Total is $19 * 8.66 = 164.5$

But I suspect that at most the loss in points for not doing this would be 0.5 for 10 total on the problem

To Part c) based on conditional expectation

$$H_0: u_{400,1986} = u_{400,2006}$$

$$H_A: u_{400,1986} > u_{400,2006} \text{ One Sided Alternative.}$$

Choose Type 1 Error of 0.05

90 in 1986 is $>$ 75 in 2006 so result is in the direction of the alternative

$(75 + 90) / 2 = 82.5$ so under H_0 the expected number in both years is 82.5

$$\text{Test Statistic is } \sum \frac{(E - O)^2}{E} = \frac{(90 - 82.5)^2}{82.5} + \frac{(75 - 82.5)^2}{82.5} = 1.36$$

Compare to $\chi^2_1(1 - 2 \times 0.05) = 2.72$. $1.36 < 2.72$ so one sided $P > 0.05$

SPRING 2012

- 1) A researcher wants to compare two methods (M1 vs M2) to teach math. The researcher picks a random sample of 8 students, and assigns at random 4 students to learn under M1, and the other 4 students under M2. At the end of the term, the researcher picks at random 3 algebra problems, and 3 geometry problems. All 8 students are given the same 6 problems. The response variable Y is the number of minutes to solve the problem. In what follows, let M denote Method, S denote Subject, P denote Problem, and T denote Type of math(algebra vs geometry).

Competency: Design of experiments – Balanced Design

[8 Students (S) Nested within 2 Methods (M)] X [6 Problems (P) Nested within 2 Types(T)]
Total df = 8 subjects x 6 problems - 1 = 48 - 1 = 47

THE PROBLEM DOES NOT MENTION INTERACTION WHICH MEANS THAT YOU CAN GO EITHER WAY. I THINK THAT IT IS SIMPLER TO ASSUME NO INTERACTION

- a) Write out the ANOVA table just showing Sources of Variability and degrees of Freedom. and indicate which SV are fixed effects and which are random.

ANOVA Table

	SS	df	MS	
M	SSM	1	SSM/1	This is a Fixed Effect
S(M)	SSS	6	SSS/6	This is a Random Effect
T	SST	1	SST/1	This is a Fixed Effect
P(T)	SSP	4	SSP/4	This is a Random Effect
Error	SSE	35	SSE/35	This is technically also a Random Effect
Total	SSA	47		

Where $\sum(.)$ denotes summations made over all 48 observations

$$SSS = \sum (\bar{Y}_{ms..} - \bar{Y}_{m...})^2 \text{ (or } = 6 * \sum_m \sum_s (\bar{Y}_{ms..} - \bar{Y}_{m...})^2 \text{)}$$

$$SSM = \sum (\bar{Y}_{m...} - \bar{Y}_{....})^2 \text{ (or } = 24 * \sum_m (\bar{Y}_{m...} - \bar{Y}_{....})^2 \text{)}$$

$$SSP = \sum (\bar{Y}_{...tp} - \bar{Y}_{...t.})^2 \text{ (or } = 8 * \sum_t \sum_p (\bar{Y}_{...tp} - \bar{Y}_{...t.})^2 \text{)}$$

$$SST = \sum (\bar{Y}_{...t.} - \bar{Y}_{....})^2 \text{ (or } = 24 * \sum_t (\bar{Y}_{...t.} - \bar{Y}_{....})^2 \text{)}$$

$$SSE = \sum (Y_{mstp} - \bar{Y}_{ms..} - \bar{Y}_{..tp} + \bar{Y}_{....})^2 \text{ (or } \sum_m \sum_s \sum_t \sum_p (Y_{mstp} - \bar{Y}_{ms..} - \bar{Y}_{..tp} + \bar{Y}_{....})^2 \text{)}$$

$$SSA = \sum (Y_{mstp} - \bar{Y}_{....})^2 \text{ (or } \sum_m \sum_s \sum_t \sum_p (Y_{mstp} - \bar{Y}_{....})^2 \text{)}$$

b) To test for Method effect, what MS terms go in numerator and what MS terms go in denominator.

Numerator = SSM / 1

Denominator = SSS / 6

2) Scientists are studying the sizes of radio-electric disturbances known as “ka-chings” that occur on Mars and Pluto. They have obtained 15 measures of ka-chings from Mars and 10 measures from Pluto. Since Mars is closer to Earth, ka-chings of any size on Mars can be detected. However, because Pluto is much further away from Earth, only ka-chings that are larger than 3 units can be measured.

Table of the 15 ordered Ka-ching measures from Mars and 10 ordered measures from Pluto

Mars	0.1	0.3	0.5	0.7	1	20	200	600	800	10,000	100,000	250,000	500,000	2,000,000
Pluto	5	50	55	1,200	2,000	5,000	25,000	40,000	1,000,000	10,000,000				

- A. It is believed that the median size of a Ka-ching on Mars is 6 units. Statistically test if the data dispute this belief with a Type-1 error of 0.05.

Competency – Nonparametric Statistics (Sign Test)

- Properties of a binomial

ANSWER - 5 of 15 observations were less than 6. If the median was 6, then the probability for this is.

$$\left[\binom{15}{0} + \binom{15}{1} + \binom{15}{2} + \binom{15}{3} + \binom{15}{4} + \binom{15}{5} \right] * (0.5)^{15} = (1 + 15 + 105 + 455 + 1365 + 3003) * (0.5)^{15} \\ = 0.151$$

Converting this to a two sided test gives $P = 2 * 0.151 = 0.302$

So the chances of seeing 5 Ka-chings from Mars less than 6 and 15 greater than 6 if the median is 6 is 0.302. The data do not refute the belief

- B. Perform the best test within the limits of the available data of whether the central tendency of the size of a Ka-ching is the same for Pluto as for Mars. Remember, that because Pluto is much further away from Earth, only Ka-chings that are larger than 3 units can be measured on Pluto. Again use a type-1 error of 0.05.

COMPETENCIES

- Understanding of conditioning (must be > 3)
- Understanding Assumptions of Central Limit Theorem, data not normal
- Knowledge of either non-parametric comparison of two samples or of using the logarithmic transformation to normalize data

Since Ka-Chings ≤ 3 cannot be measured on Pluto, we must restrict our comparisons to Ka-Chings > 3 on both planets. The null hypothesis then becomes that given Ka-Chings are > 3 , then they are the same size on both planets.

With Ka-chings ≤ 3 on Mars excluded, we have the following data

Mars	20	200	600	800	10,000	100,000	250,000	500,000	2,000,000	5,000,000
Pluto	5	50	55	1,200	2,000	5,000	25,000	40,000	1,000,000	10,000,000

These data are clearly not normally distributed so the rank sum test should be used.
The Ranks added n are

	2	5	6	7	11	14	15	16	18	19
Mars	20	200	600	800	10,000	100,000	250,000	500,000	2,000,000	5,000,000
Pluto	5	50	55	1,200	2,000	5,000	25,000	40,000	1,000,000	10,000,000
	1	3	4	8	9	10	12	13	17	20

There are no ties so no correction for ties is need.

The sum of the ranks for Mars is $U_1 = 2 + 5 + 6 + 7 + 11 + 14 + 15 + 16 + 18 + 19 = 113$

Since $n_1 = 10$ and $n_2 = 10$ both n_1 and n_2 are ≥ 10

And by normal approximation with no correction for ties

$$U_1 \sim N [n_1 (n_1 + n_2 + 1)/2, n_1 n_2 (n_1 + n_2 + 1)/12]$$

$$\text{or } N [10 (21)/2, (10)(10) (21)/12] \text{ or } N(105, 175)$$

$$\text{so the Z-test is } \frac{113 - 105}{\sqrt{175}} = \frac{8}{13.23} = 0.60 \quad P > 0.05$$

so conditioning on the Ka-Ching being 3 or greater, there is no statistical evidence for directional difference in the central tendency of size between Pluto and Mars.

WORKING THIS OUT USING THE LOG TRANSFORMATION

Log Transformed Observations

Mars	1.301	2.301	2.778	2.903	4.000	5.000	5.398	5.699	6.301	6.699
Pluto	0.699	1.699	1.740	3.079	3.301	3.699	4.398	4.602	6.000	7.000

Summary Statistics

	Mean	Std-Dev
Mars	4.238	1.848

NOTE for both Groups Mean ~ Median suggesting normal approximation will be good

Pluto	3.622	1.967
-------	-------	-------

$$T\text{-test} = \frac{4.328 - 3.622}{\sqrt{\frac{9 * (1.848)^2 + 9 * (1.967)^2}{9 + 9} \left(\frac{1}{10} + \frac{1}{10}\right)}} = \frac{0.706}{\sqrt{3.64(0.2)}} = 0.827 \text{ two sided p for } t = 18$$

df = 0.48

So again conditioning on the Ka-Ching being 3 or greater, there is no statistical evidence for directional difference in the central tendency of size between Pluto and Mars.

NOTE - If in a problem such as this you have the train of thought right but the numbers or wrong or you write in “Ran Out of Time” you should get some or maybe even most of the points.

FALL 2011

3. We have the following model for the probability that a pine tree will survive

$P_{Survival} | F = \frac{e^{\alpha + \beta F}}{1 + e^{\alpha + \beta F}}$ where α and β are unknown parameters and F is the number of units of fertilizer given to the tree. The following MLE point estimates and standard deviations are obtained for α and β ; $\hat{\alpha} = 0.12 \pm 0.01$, $\hat{\beta} = 0.024 \pm 0.003$ with $Corr(\hat{\alpha}, \hat{\beta}) = -0.1$

- A. Give a 95% confidence interval for the ratio of odds of survival for a tree given $F=100$ units over those of a tree given $F=80$ units of fertilizer

COMPETENCIES

- Logistic Regression

$$\begin{aligned} e^{[\hat{\beta} - 1.96(\text{stderr } \beta)](100-80)} &< OR(100 \text{ Vs } 80) < e^{[\hat{\beta} + 1.96(\text{stderr } \beta)](100-80)} \\ e^{[0.024 - 1.96(0.003)](100-80)} &< OR(100 \text{ Vs } 80) < e^{[0.024 + 1.96(0.003)](100-80)} \\ 1.43 &< OR(100 \text{ Vs } 80) < 1.81 \end{aligned}$$

It costs \$1000 for each tree that dies. Let C denote the cost per unit of fertilizer given to the tree. Find the MLE for the value of C at which $F=80$ units and $F=100$ have the same expected combined cost for fertilizer and dead trees.

COMPETENCIES

- Logistic Regression
- Problem solving with Algebra

TOTAL COST IS

$$1,000 * [\text{Probability Tree Dies Given } C] + [C * \text{Units of Fertilizer}]$$

$1 - \frac{e^{\alpha + \beta F}}{1 + e^{\alpha + \beta F}}$ is the probability a tree dies so we solve for C which gives equal net costs at 80 and 100 units of fertilizer

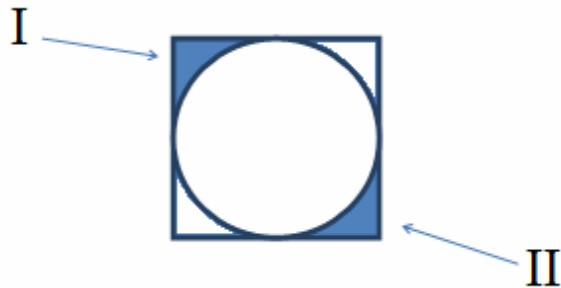
$$\left[1,000 \left(1 - \frac{e^{0.12 + 80 \cdot 0.024}}{1 + e^{0.12 + 80 \cdot 0.024}}\right) + C80\right] = \left[1,000 \left(1 - \frac{e^{0.12 + 100 \cdot 0.024}}{1 + e^{0.12 + 100 \cdot 0.024}}\right) + C100\right] \text{ or}$$

$$\left[1,000 \frac{1}{1 + e^{0.12 + 80 \cdot 0.024}}\right] = \left[1,000 \frac{1}{1 + e^{0.12 + 100 \cdot 0.024}} + C20\right] \text{ or}$$

$$C = 50 \left[\frac{1}{1 + e^{0.12 + 80 \cdot 0.024}} - \frac{1}{1 + e^{0.12 + 100 \cdot 0.024}} \right] = 50[0.1151 - 0.0745] = 2.03$$

SPRING 2011

4. A rocket company is firing missile into a square region with each side of length of $2r$ that has an inscribed circle with a radius r as shown below.



One of the scientists believes that the rockets are biased to go towards the upper left and away from the lower right. Specifically that the rocket is more likely to land in Region I than it is to land in Region II.

- A. State the scientist's Null and Alternative Hypothesis and all the assumptions needed to test these.

COMPETENCY

- Formulating a Hypothesis Test

H_0 – Probability to fall in region I is the same as probability to fall in region II (and the probability to fall in any region of the square is proportional to the size of the region)

H_A – Probability to fall in region I is greater than probability to fall in region II

Assumptions all rocket shots have independent and identical probabilities to fall at any point in the square

- B. What are the expected numbers of rockets each that land in Region I and in Region II if H_0 is true

COMPETENCY

- Algebraic Construction of a Probability based on areas of a square and circle

Ratio of the area outside of circle to that in square is $[(2r)^2 - \pi r^2] / (2r)^2 = [4 - \pi] / 4 = 0.2146$ but I and II are $1/4^{\text{th}}$ of this area or $0.2146/4 = 0.05365$ of the entire square so if H_0 is true $0.05365 * 1,000 \sim 53.5$ rockets will land in each region.

- C. Out of 1,000 shots, fired, 63 have landed in Region I and 44 have landed in Region II.
Give a best large sample statistical test of the Null Hypothesis against the Alternative
at $\alpha = 0.05$

COMPENTENCIES

- Large sample approximation to the multinomial

Or

- Conditioning and large sample approximation to the binomial

CONDITIONAL APPROACH - Since Region I and Region II have the same size, condition on the sum of shots falling in either regions

$$44 + 63 = 107.$$

Given that 107 fell in either regions, the probability for each of these to fall in I is 0.5 so the large sample distribution of the number of shots falling in Region I is

$$N(107/2, \sqrt{107 * 0.5 * 0.5}) = N(53.5, 5.17)$$

$$\text{The Test Statistic is } \frac{63 - 53.5}{5.17} = 1.84$$

Since $1.84 > 1.65$ reject H_0 with a large sample one sided P-value of 0.05

UNCONDITIONAL APPROACH

The probability to fall in Region I is $0.25 * (4 - 3.14)/4 = 0.05375$
and has variance $0.05375 * (1 - 0.05375)$

The probability to fall in Region II is $0.25 * (4 - 3.14)/4 = 0.05375$
and has variance $0.05375 * (1 - 0.05375)$

The covariance of falling in Regions I and II is $-(0.05375)^2$

So under H_0 with 1,000 shots fired, the difference in the numbers falling in Region I and II has a large sample distribution of

$$N(0, \sqrt{(1,000)[0.05375(1 - 0.05375) + 0.05375(1 - 0.05375) - 2 * -(0.05375)^2])$$

Or $N(0, 10.37)$

So the Test Statistic is $\frac{63 - 44}{10.37} = 1.84$. Since $1.84 > 1.65$ Reject H_0 at a one sided $P = 0.05$

FALL 2010

1. The Body mass Index BMI of a person is $\text{Weight} / (\text{Height})^2$ where weight is in Kg and height is in meters. BMI is used as a measure of height adjusted body mass.

A. Suppose for country X, all of the following are known about height and BMI

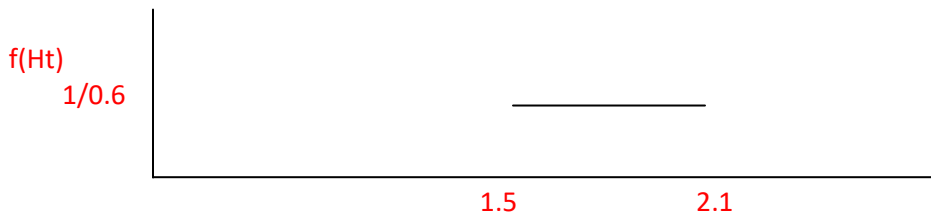
- * height of adults is uniformly distributed on 1.5 to 2.1 Meters
- * height is independent of BMI
- * the true population mean for BMI is 23.0

what is the mean weight of an adult in country X?

COMPETENCIES

Understanding of what Expectation is
Understanding of Properties of Expectations
Calculus

ANSWER



- $\text{BMI} = \text{Weight} / (\text{Height})^2$ means $\text{Weight} = (\text{Height})^2 * \text{BMI}$
- The expected value of $(\text{Height})^2$ is

$$\int_{1.5}^{2.1} \frac{1}{0.6} (X^2) dx = \left| \frac{1}{0.6} \frac{X^3}{3} \right|_{1.5}^{2.1} = \frac{1}{1.8} (9.261 - 3.375) = 3.27$$

- Since Height and BMI are independent it follows that $E[\text{Weight}] = E[(\text{Height})^2 * \text{BMI}] = E[(\text{Height})^2] * E[\text{BMI}] = 3.27 * 23.0 = 75.21$

- B. But now someone questions whether height and BMI are independent in this population and has collected the following data for means and standard deviations of BMI for different ranges of height

Height Category	Mean BMI
1.5 – 1.65 = 20 subjects	24
1.65 – 1.80 n= 20 subjects	22
1.80 – 1.95 n=20 subjects	26
1.95-2.10 n=20 subjects	20

The standard deviation of BMI is the same for all height categories and the pooled estimate of the variance (with $80 - 4 = 76$ d.f.) is 56.5

Find the best test (which most efficiently adjusts for multiple comparisons) of

$$H_0: \mu_{1.5-1.85} = \mu_{1.65-1.80} = \mu_{1.80-1.95} = \mu_{1.55-2.10}$$

Against

H_a : At least one of the group means is different

NOTE – The test is for any group difference and not for a trend

COMPETENCIES – Linear Models, Multiple Comparisons

ANSWER

I would use Tukey's Range (or studentized t) rather than the one-way ANOVA F-Test

Pooled Variance is = 56.5

$$\text{So } \frac{MAX - MIN}{SEM} = \frac{26 - 20}{\sqrt{56.5/20}} = \frac{6}{1.81} = 3.56$$

and $3.56 < 3.69$ the threshold standardized value to reject H_0 at $\alpha=0.05$ for 4 groups with 120 d.f. (which is more than the 76 df we have here. So we fail to reject H_0)

But I would give most (maybe all) of the maximum possible points for solutions done using either the F-Test one way ANOVA

3. An experiment was designed where 3 machines are randomly selected and each machine is assigned to a different group of 3 operators selected at random. Each operator **uses the machine 2 times at different periods during a given week**. The amounts of time needed to produce 1 item given in standard units are below:

Machine	1			2			3		
Operator	1	2	3	4	5	6	7	8	9
	13.2	14.1	13.8	9.5	12.6	9.7	17.4	16.0	15.4
	14.3	14.6	12.7	9.8	11.7	11.2	17.6	13.0	14.4

- (a) Calculate the appropriate quantities (SS df MS F) and Write down the standard ANOVA table.
- (b) Describe how to test if there are differences in completion time among the machines.
- (c) Describe how to test if there are differences in completion times between operators within machines.

COMPETENCY

- Linear Models
- Design of Experiments
- Nested Models

Since the time periods are not labeled 1 and 2 you can assume that these are just random observations and that order of the observation does not matter

ANSWER TO (a)

Nested Model with Operator Nested Within Machine. Two measures per operator

Overall Mean

$$\bar{y}_{...} = 13.39$$

Machine Means

$$\bar{y}_{1..} = 13.78$$

$$\bar{y}_{2..} = 10.75$$

$$\bar{y}_{3..} = 15.63$$

Operator Means

$$\bar{y}_{11.} = 13.75$$

$$\bar{y}_{12.} = 14.35$$

$$\bar{y}_{13.} = 13.25$$

$$\bar{y}_{21.} = 9.65$$

$$\bar{y}_{22.} = 12.15$$

$$\bar{y}_{23.} = 10.45$$

$$\bar{y}_{31.} = 17.5$$

$$\bar{y}_{32.} = 14.5$$

$$\bar{y}_{33.} = 14.9$$

Due to time constraints of the exam it helps to know orthogonal decompositions calculation of these variances about a mean

$$(a) \quad SS_T = \sum_{i=1}^3 \sum_{j=1}^3 \sum_{k=1}^2 y_{ijk}^2 - 3 \times 3 \times 2 (\bar{y}_{...}^2) = 3325.94 - 3226.722 = 99.22$$

$$SS_{Machine} = 2 \times 3 \sum_{i=1}^3 \bar{y}_{i..}^2 - 3 \times 3 \times 2 (\bar{y}_{...}^2) = 3299.663 - 3226.722 = 72.94$$

$$SS_{op(machine)} = 2 \times \sum_{i=1}^3 \sum_{j=1}^3 \bar{y}_{ij.}^2 - 2 \times 3 \times \sum_{i=1}^3 \bar{y}_{i..}^2 = 3318.01 - 3299.663 = 18.35$$

$$SS_E = SS_T - (SS_{Machine} + SS_{op(machine)}) = 99.22 - (72.94 + 18.35) = 7.93$$

ANOVA Table

	SS	df	MS	F
Machine	72.94	2	36.47	11.92 (36.47 / 3.06)
Op(Machine)	18.35	6	3.06	3.48 (3.06 / 0.88)
Error	7.93	9	0.88	
Total	99.22	17		

ANSWER TO (b)

Compare 11.92 with $F_{\alpha,2,6}$. Reject H_0 if $11.92 > F_{\alpha,2,6}$

ANSWER TO (c)

Compare 3.48 with $F_{\alpha,6,9}$. Reject H_0 if $3.48 > F_{\alpha,6,9}$

SPRING 2010

1. The Kenton Company is testing 4 different package designs for a new breakfast cereal; 20 stores, with about equal sales volumes, are the experimental units. Each store is randomly assigned one package designs, with each package design being assigned to 5 stores. A fire later destroys one store so this store is dropped from the study. Sales, in number of cases, for the study period are recorded in Table 1. The characteristics of 4 package designs are shown in Table 2. The data is analyzed using one-way layout with fixed effects.

Table 1: Number of Cases Sold by Stores for Each of Four Package Designs

	Package Design			
	1	2	3	4
Cases Sold Per Store	11	12	23	27
	17	10	20	33
	16	15	18	22
	14	19	17	26
	15	11		28
Mean	$\bar{y}_1=14.6$	$\bar{y}_2=13.4$	$\bar{y}_3=19.5$	$\bar{y}_4=27.2$
Standard Deviation	2.3	3.6	2.6	4.0

Table 2: Characteristics of Four Package Designs

Package Design	Characteristics
1	3 color, with cartoons
2	3 color, without cartoons
3	5 color, with cartoons
4	5 color, with cartoons

COMPETENCIES

- One Way ANOVA, unbalanced design

A. Copy the ANOVA table below to your Blue Book and fill in the blanks

One-way ANOVA Table

Source	Degrees of Freedom	Sum of Squares	Mean Squares	F
Between design	3	_____	_____	_____
Error	_____	_____	_____	
Total	18	_____		

A One-way ANOVA Table

Source	Degrees of Freedom	Sum of Squares	Mean Squares	F
Between design	3	<u>610.5</u>	<u>203.5</u>	<u>19.4</u>
Error	<u>15</u>	<u>157.2</u>	<u>10.5</u>	

$$\bar{y} = \frac{5\bar{y}_1 + 5\bar{y}_2 + 4\bar{y}_3 + 5\bar{y}_4}{5+5+4+5} = \frac{5(14.6) + 5(13.4) + 4(19.5) + 5(27.2)}{5+5+4+5} = 18.6$$

$$\begin{aligned} SSE &= 4 (sd_{y_1})^2 + 4 (sd_{y_2})^2 + 3 (sd_{y_3})^2 + 4(sd_{y_4})^2 \\ &= 4 (2.3)^2 + 4 (3.6)^2 + 3 (2.6)^2 + 4(4.0)^2 = 157.2 \end{aligned}$$

$$\begin{aligned} SSD &= 5(\bar{y}_1)^2 + 5(\bar{y}_2)^2 + 4(\bar{y}_3)^2 + 5(\bar{y}_4)^2 - 19(\bar{y})^2 = \\ &= 5(14.6)^2 + 5(13.4)^2 + 4(19.5)^2 + 5(27.2)^2 - 19(18.6)^2 = 7183.8 - 6573.2 = 610.5 \end{aligned}$$

B. Conduct an F test to test at $\alpha = 0.05$ of the null hypothesis that the four package designs have the same sales.

$F_{3,15,0.05}=3.29$, $F=19.4= 203.5 / 10.5 > F_{3,15,0.05}$. Reject the null hypothesis.

COMPETENCIES

- Multiple Comparisons, unequal sample size

C. Use Tukey's method to perform multiple comparisons of the four package designs at $\alpha = 0.05$

$$\hat{\sigma}^2 = 10.5 \quad \text{so} \quad \sqrt{\hat{\sigma}^2 \left(\frac{1}{5} + \frac{1}{5} \right)} = 2.05, \quad \sqrt{\hat{\sigma}^2 \left(\frac{1}{5} + \frac{1}{4} \right)} = 2.17, \quad \frac{1}{\sqrt{2}} q_{4,15,0.05} = \frac{1}{\sqrt{2}} 4.08 = 2.88$$

$$t_{1,2} = \frac{\bar{y}_{1\bullet} - \bar{y}_{2\bullet}}{2.05} = 0.59, \quad t_{1,3} = \frac{\bar{y}_{1\bullet} - \bar{y}_{3\bullet}}{2.17} = -2.26$$

$$t_{1,4} = \frac{\bar{y}_{1\bullet} - \bar{y}_{4\bullet}}{2.05} = -6.15, \quad t_{2,3} = \frac{\bar{y}_{2\bullet} - \bar{y}_{3\bullet}}{2.17} = -2.81$$

$$t_{2,4} = \frac{\bar{y}_{2\bullet} - \bar{y}_{4\bullet}}{2.05} = -6.73, \quad t_{3,4} = \frac{\bar{y}_{3\bullet} - \bar{y}_{4\bullet}}{2.17} = -3.55$$

$|t_{1,4}|, |t_{2,4}|$, and $|t_{3,4}|$ are larger than $\frac{1}{\sqrt{2}} q_{4,15,0.05} = \frac{4.37}{1.414} = 2.88$ so conclude that design 4 sells better than all other designs

FALL 2009

1. The current laboratory test for amounts of “Testro” in a student’s blood has too much intraperson measurement variation and a new test for Testro is believed to perform better. Eight Graduate Students each are tested twice for Testro with the Standard Test and twice with the New Test with the following results.

Student	TESTRO STANDARD TEST				TESTRO NEW TEST		
	1 st Test	2 nd Test	Mean of Both Tests		1 st Test	2 nd Test	Mean of Both Tests
1	30	36	33		28	24	26
2	44	44	44		41	43	42
3	78	70	74		73	78	75.5
4	55	61	58		60	52	56
5	62	67	64.5		71	67	69
6	39	38	38.5		40	40	40
7	23	33	28		30	26	28
8	52	46	49		48	44	46

Assume that all tests on each person are normally distributed and independent and what is observed from each test j on person i , is $Y_{ij} = X_i + \epsilon_{ij}$ Where

- X_i is the True value for person i
- ϵ_{ij} is the measurement error of test j on person i which has mean 0 and variance σ_s^2 for the standard test and mean 0 with variance σ_n^2 for the “New” Test

- A. Give the Best Estimate for σ_s^2 and σ_n^2 from the data

COMPETENCY – Estimation of Variance

ANSWER

For the same person i with both tests using the standard method

$$Y_{ij} - Y_{ij'} = (X_i + \epsilon_{ij}) - (X_i + \epsilon_{ij'}) = (\epsilon_{ij} - \epsilon_{ij'}) \text{ which has expectation 0 and variance } 2\sigma_s^2$$

For the same person i with both tests using the “New” Method

$$Y_{ij} - Y_{ij'} = (X_i + \epsilon_{ij}) - (X_i + \epsilon_{ij'}) = (\epsilon_{ij} - \epsilon_{ij'}) \text{ which has expectation 0 and variance } 2\sigma_n^2$$

Thus the best estimates are:

$$\hat{\sigma}_s^2 = \frac{1}{2} \frac{(30-36)^2 + (44-44)^2 + (78-70)^2 + (55-61)^2 + (62-67)^2 + (39-38)^2 + (23-33)^2 + (52-46)^2}{8}$$

$$= 18.625$$

$$\hat{\sigma}_n^2 = \frac{1}{2} \frac{(28-24)^2 + (41-43)^2 + (73-78)^2 + (60-52)^2 + (71-67)^2 + (40-40)^2 + (30-26)^2 + (44-46)^2}{8}$$

$$= 9.0625$$

B. State any additional Assumptions needed and test

$H_0 : \sigma_s^2 = \sigma_n^2$ against $H_a : \sigma_s^2 > \sigma_n^2$ with a Type 1 Error of 0.05

COMPETENCY – F Distribution and F Test

Assume Normality then under H_0 $2\hat{\sigma}_n^2 / 2\hat{\sigma}_s^2 = \hat{\sigma}_n^2 / \hat{\sigma}_s^2 \sim F_{8,8}$

The Test Statistic is $9.0625 / 18.625 = 0.4866 > 0.291 = F_{8,8} (0.05)$ so do not reject H_0 .

C. Now only consider the **Means of both tests** as the outcome. Dr. W claims these means are **not normally distributed**. Assume Dr. W's claim and do a two sided test of the null hypothesis H_0 : that for a given person that the median of the differences between two test means is 4 units higher for the new test than for the standard test against the alternative that the median difference is something else. Use a type-1 error of 0.10.

COMPETENCY – Nonparametric, sign test

Assumption = Not Normality

Sign Test Using $D_4 = (N_i - S_i) - 4$

Person	Mean of Standard Tests	Mean of New Tests	Difference	Difference Minus 4
1	33	26	7	3
2	44	42	2	-2
3	74	75.5	-1.5	-7.5
4	58	56	2	-4
5	64.5	69	-4.5	-8.5
6	38.5	40	-1.5	-5.5
7	28	28	0	-4

8	49	46	3	-1
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Seven of the eight paired differences are < 4. The Two sided P-value for this is

$$2 * \left[\binom{8}{8} (0.5)^8 + \binom{8}{7} (0.5)^8 \right] = 0.07 \quad \text{Since } 0.07 < 0.10, \text{ reject } H_0$$

SPRING 2009

4. Two methods (Method A and Method B) to produce vials of a vaccine are tested with the outcome being that the vial is either good or bad. Each method is allowed to produce new vials one at a time until the first bad vial from that method was observed, then production using that method is stopped. In other words,

- For Methods A & B each, if the last vial produced is good the method is allowed to produce another vial.
- But once a bad vial is produced, the Method is stopped and no new vials are produced with that method.

COMPETENCIES – Negative Binomial / Geometric Distribution, means and variances, conditioning

A. Assume that the probability for a vial produced by Method A to be bad is $p = 0.05$ and the probability for a vial produced by Method B to be bad is 0.10 and all vials are independent.

- i) What is the expected total number of good vials produced by both Methods A and B combined before both methods are stopped?
- ii) What is the variance of the total number of good vials produced by both Methods A and B combined before both methods are stopped

ANSWER

A. Negative binomials ($p=.05, r=1$) and ($p=.10, r=1$)

Expectation of successes for a negative binomial is $r(1-p)/p$
 $= (1-.05)/.05$ or 19 for Method A
 $= (1-.10)/.10$ or 9 for Method B

Variance of successes is $r(1-p)/(p)^2$
 $= (1-.05)/(.05)^2 = 380$ for method A
 $= (1-.10)/(.10)^2 = 90$ for method B

Total Expectation for Both Methods = $19 + 9 = 28$

Total Variance for Both Methods = $380 + 90 = 470$

B. Now assume that the probabilities for vials produced by each method to be bad are unknown and the following results are observed:

- Method A - On its first try, the vial was bad so production stopped
- Method B produced 15 good vials but on its 16th try, the vial was bad so production was stopped then

Derive the appropriate model that conditions on 15 bad vials being produced and give a P-value for observing a result this extreme if each method is equally likely to produce a bad vial.

Negative binomial models with possibly different parameters p_A and p_B

For Method A if X is the number of successes observed before the first failure

$$P\{X=x\} = p_A (1-p_A)^x$$

For Method B if Y is the number of successes observed before the first failure

$$P\{Y=y\} = p_B (1-p_B)^y$$

Let $Z = X + Y$ be the number of successes total observed before the single failure in each group or 2 failures in both groups

Under the null hypothesis that $p_A = p_B = p$ WLOG assume that Method A was applied first then Method B, the distribution of Z is

$$P\{Z=z\} = \binom{z+1}{1} p^2 (1-p)^z$$

This means that given each method has the same probability to produce an impure lot, then conditional on 15 successes being observed in both Method A and B combined before their first failures, the probability to observe 0 successes in Method A before its failure and 15 successes in Method B before its failure is:

$$P\{C|D\} = P\{C \cap D\} / P\{D\}$$

Where

D = 15 successes before 2 failures in both methods under H_0

C = 0 successes before 1 failure in method A and 15 successes before 1 failure in Method B under H_0

=

$$\frac{p (1-p)^x p (1-p)^y}{\binom{z+1}{1} p^2 (1-p)^z} = \frac{p (1-p)^0 p (1-p)^{15}}{\binom{16}{1} p^2 (1-p)^{15}} = \frac{1}{16}$$

Thus for a two sided test, the P-Value is $2/16 = 1/8$

FALL 2008

2. The time it takes for an engine to fail is exponentially distributed. We have observed the following ordered failure times (in years) for 8 engines

1.2 2.3 4.3 5.4 7.6 8.4 10.2 14.5

COMPETENCIES – Exponential Distribution. Confidence limits

- A. Give the best estimate of the mean failure time and the standard deviation of this estimate and state all assumptions needed

The additional assumptions beyond those stated in the problem are that the observations are independent, identical.

For an exponential distribution, the mean failure time is λ and the MLE for λ is

$$\hat{\lambda} = \bar{X} = [(1.2 + 2.3 + 4.3 + 5.4 + 7.6 + 8.4 + 10.2 + 14.5) / 8] \\ = 6.74$$

and the MLE for the std-dev of $\hat{\lambda}$ is $\hat{\lambda} / \sqrt{n} = 6.74 / 2.828 = 2.38$

- B. Give the best estimate of the timepoint X for which exactly 90% of the engines fail in X or less years. Give a 95% confidence interval for X

Ninety percent of the engines fail by time x where $e^{-x/\lambda} = (1 - 0.90) = 0.10$

Substituting $\hat{\lambda}$ for λ gives $e^{-x/\hat{\lambda}} = e^{-x/6.74} = 0.10$

Working backwards

$$e^{-x/6.74} = 0.10 \\ \log[e^{-x/6.74}] = \log[0.10] \\ -x/6.74 = -2.3025 \\ x = 2.3025 * 6.74 = 15.5$$

To get 95% confidence intervals for X, plug in the corresponding 95% confidence intervals for λ

from $\hat{\lambda} \pm 1.96 \hat{\lambda} / \sqrt{n}$

or

$$\lambda_L = 6.74 - 1.96 * 2.38 = 2.08$$

$$\lambda_U = 6.74 + 1.96 * 2.38 = 11.40$$

So the

$$\text{Lower Limit is } x = 2.3025 * 2.08 = 4.79$$

$$\text{Upper Limit is } x = 2.3025 * 11.40 = 26.24$$

So the 95% CI is $4.79 < X < 26.24$